

AP CSP | Lesson 05: Sampling & Representation

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: C

C is correct: $32 + 8 + 4 = 44$. A omits the 4 position. B uses 2 instead of 4. D uses 16 instead of $8 + 4$. The stated integer interpretation is necessary; the same bit pattern could mean something else under another encoding.

Q2. Answer: C

C is correct: samples becomes 8, then bits becomes $8 \times 3 = 24$. A uses the old samples value of 4. B follows the incorrect expression $(4 + 3) \times 2$. D follows 8×4 rather than the final instruction. Assignment statements execute in the stated order.

Q3. Answer: D

D is correct: the rate changes from 0.5 measurements per second to 2 per second, a factor of four. A reverses the comparison by using intervals as though they were rates. B gives the new numerical rate but not the factor of change. C overlooks the changed interval.

Q4. Answer: C

C is correct: 5.2 is 0.8 from 6, less than its distance from any other allowed level. A is 2.2 away; D is 3.8 away. B is the original measurement but is not one of the allowed stored values. The stored result is an approximation.

Q5. Answer: A and C

A and C each double sample count $\times$ bits per sample. B changes the minimum code length from 3 to 4 bits, giving a factor of $4/3$ if minimum lengths are used, not two; code count and bit count differ. D halves the number of samples, so it halves payload. Award one mark only for A and C, with no other selections.

Q6. Answer: B

B is correct: $10 \times 6 \times 4 = 240$ bits; $240 / 8 = 30$ bytes. A can result from 6×4 , which omits the width and the bit-to-byte conversion. C is the number of pixels, not bytes. D gives the bit count but labels it bytes.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: (a) The smooth curve is an analog model; the stored file is digital. (b) The file preserves selected information about the temperature while omitting other detail. (c) It may omit changes between hourly measurements and differences smaller than the chosen value precision.

1 mark: Correctly classifies both the smooth curve and the stored file.

1 mark: Explains abstraction as representing selected useful information while omitting detail.

1 mark: Identifies changes between measurement times as potentially omitted.

1 mark: Identifies finer value differences removed by rounding as potentially omitted.

Q8. 4 marks

Answer: (a) 0, 3, 6, 9. (b) Two bits: $2^2 = 4$ codes. (c) $4 \times 2 = 8$ bits, or one byte.

1 mark: Correctly maps the first two measurements to 0 and 3.

1 mark: Correctly maps the last two measurements to 6 and 9.

1 mark: Gives two bits per sample and connects this to four possible codes.

1 mark: Calculates 8 bits or one byte for the four samples.

Q9. 4 marks

Answer: (a) Increase the sample count. (b) Increase the number of value levels. (c) Three bits per sample because $2^3 = 8$; $16 \times 3 = 48$ payload bits.

1 mark: Identifies increased sample count for time detail.

1 mark: Identifies increased value levels for nearby value distinctions.

1 mark: Gives a minimum of three bits per sample for eight levels.

1 mark: Calculates $16 \times 3 = 48$ bits.

Q10. 4 marks

Answer: (a) BA C. (b) 10 00. (c) No: it needs the character mapping and grouping rules. (d) No. This is a custom four-symbol code; other text encodings can use different code lengths and symbol sets.

1 mark: Decodes BA C, with a space between A and C.

1 mark: Encodes CA as 10 00 or 1000.

1 mark: Explains that reliable interpretation needs the agreed mapping or grouping rules.

1 mark: Rejects the universal two-bit claim and identifies the code as a custom limited-symbol example.

Q11. 4 marks

Answer: (a) $20 \times 12 = 240$ pixels. (b) $240 \times 2 = 480$ bits = 60 bytes. (c) A factor of four: twice the width $\times$ twice the height.

1 mark: Calculates 240 pixels.

1 mark: Calculates 480 bits.

1 mark: Converts correctly to 60 bytes.

1 mark: Gives a factor of four and supports it with both dimensions doubling.

Q12. 4 marks

Answer: (a) More measurement times may capture detail missed between widely spaced samples. (b) A finite record still omits unsampled values and uses finite precision. (c) Payload doubles because the sample count doubles while bits per sample stay fixed. (d) Increase the number of available value levels, using enough bits to encode them.

1 mark: Explains that more measurement times may capture additional time detail.

1 mark: Rejects guaranteed exactness using finite sampling or finite value precision.

1 mark: States and justifies that uncompressed sample payload doubles.

1 mark: Identifies more value levels, with sufficient code capacity, as the relevant different setting.